Detailed Solution of the Problem Using the Two-Phase Simplex Method

Original Problem

Minimize:

$$Z = x_1 + x_2 + x_3$$

Subject to:

$$\begin{cases} 2x_1 + x_2 - x_3 \leq 10 \\ x_1 + x_2 + 2x_3 \geq 20 \\ 2x_1 + x_2 + 3x_3 = 60 \\ x_1, x_2, x_3 \geq 0 \end{cases}$$

Phase I: Elimination of Artificial Variables

Step 1: Convert inequalities to equalities

- For the $\leq$ constraint, add slack variable $s_1 \geq 0$:$2x_1 + x_2 - x_3 + s_1 = 10$
- For the $\geq$ constraint, subtract excess variable $s_2 \geq 0$ and add artificial variable $a_1 \geq 0$: $x_1 + x_2 + 2x_3 - s_2 + a_1 = 20$
- For the equality constraint, add artificial variable $a_2 \geq 0$:$2x_1 + x_2 + 3x_3 + a_2 = 60$

Step 2: Auxiliary objective function for Phase I

$$\min W = a_1 + a_2$$

Initial Tableau

Base	x_1	x_2	x_3	s_1	s_2	a_1	a_2	RHS

s_1	2	1	-1	1	0	0	0	10
a_1	1	1	2	0	-1	1	0	20
a_2	2	1	3	0	0	0	1	60
W	-3	-2	-5	0	1	0	0	80

Note: The W row is obtained by substituting a_1 and a_2 from the constraints.

Step 3: Iteration 1 — Enter x_3, Leave a_1

- **Pivot element:** 2 (row a_1, column x_3).
- **Row operations:** Divide row a_1 by 2 and update other rows to zero out column x_3.

Tableau after Iteration 1

Base	x_1	x_2	x_3	s_1	s_2	a_1	a_2	**RHS**
s_1	2.5	1.5	0	1	-0.5	0.5	0	20
x_3	0.5	0.5	1	0	-0.5	0.5	0	10
a_2	0.5	-0.5	0	0	1.5	-1.5	1	30
W	-0.5	0.5	0	0	-1.5	2.5	0	130

Step 4: Iteration 2 — Enter s_2, Leave a_2

- **Pivot element:** 1.5 (row a_2, column s_2).
- **Row operations:** Divide row a_2 by 1.5 and update other rows to zero out column s_2.

Tableau after Iteration 2

Base	x_1	x_2	x_3	s_1	s_2	a_1	a_2	RHS
$\mathbf{s_1}$	8/3	4/3	0	1	0	0	1/3	30
$\mathbf{x_3}$	2/3	1/3	1	0	0	0	1/3	20
$\mathbf{s_2}$	1/3	-1/3	0	0	1	-1	2/3	20
W	0	0	0	0	0	1	1	0

⚠️ Attention: If Phase I ends with $W \neq 0$

If at the end of Phase I the auxiliary objective function W **is not zero**, i.e., $W > 0$, this indicates that **there is no feasible solution to the original problem**.

This happens because artificial variables could not be removed from the basis, signaling inconsistency in the original constraints.

In this case, the Two-Phase Simplex method **stops and reports the problem as infeasible**.

Phase II: Optimization of the Original Objective Function

Initial Tableau Phase II

Base	x_1	x_2	x_3	s_1	s_2	RHS
$\mathbf{s_1}$	8/3	4/3	0	1	0	30
$\mathbf{x_3}$	2/3	1/3	1	0	0	20
$\mathbf{s_2}$	1/3	-1/3	0	0	1	20

Step 5: Write the original objective function in terms of the variables

$$x_3 = 20 - \frac{2}{3}x_1 - \frac{1}{3}x_2$$

$$Z = x_1 + x_2 + x_3 = x_1 + x_2 + 20 - \frac{2}{3}x_1 - \frac{1}{3}x_2 = 20 + \frac{1}{3}x_1 + \frac{2}{3}x_2$$

Step 6: Optimality Check

The coefficients of Z for the non-basic variables x_1 and x_2 are positive $\left(\frac{1}{3} \text{ and } \frac{2}{3}\right)$, indicating the current solution is optimal.

Optimal Solution

$$(x_1, x_2, x_3) = (0, 0, 20) \text{ with } Z_{\min} = 20$$

Constraint Verification

$$\begin{cases} 2(0) + 0 - 20 = -20 \leq 10 \checkmark \\ 0 + 0 + 2(20) = 40 \geq 20 \checkmark \\ 2(0) + 0 + 3(20) = 60 = 60 \checkmark \end{cases}$$

Summary

- The Two-Phase Simplex method was applied correctly.
- Phase I eliminated artificial variables with $W = 0$, indicating the problem is **feasible**.
- Phase II optimized the original objective function, finding the optimal solution.
- If $W \neq 0$ at the end of Phase I, the problem would be **infeasible** (no feasible solution).